LINUX + GIT QUICK CHEAT SHEET

A4 • Student-friendly command reference • Keep this beside your terminal

1 • BASIC LINUX COMMANDS

COMMAND	SAMPLE USAGE	WHAT IT DOES
pwd	pwd	Show the directory you are currently in
ls	ls -la	List files, including hidden files
cd	cd projects	Move into another directory
mkdir	mkdir app	Create a new directory
touch	touch notes.txt	Create an empty file
cp	cp a.txt backup.txt	Copy a file or directory
mv	mv old.txt new.txt	Move or rename a file
rm	rm notes.txt	Delete a file
cat	cat notes.txt	Display file contents
grep	grep "error" log.txt	Search for matching text
find	find . -name "*.py"	Find files using a pattern
chmod	chmod +x script.sh	Change file permissions

2 • BASIC GIT COMMANDS

COMMAND	SAMPLE	DESCRIPTION	COMMON FLAGS
git init	git init	Create/start a local Git repository	--bare
git status	git status	Show current branch and changed/staged files	-s, --short
git add	git add .	Put selected changes into staging	-A, -p
git commit	git commit -m "Add login"	Save staged changes as a commit	-m, -a
git log	git log --oneline	View previous commits	--oneline, -p, --graph
git diff	git diff	Compare working files with staging	--cached, HEAD
git restore	git restore file.txt	Restore/discard file changes	--staged, --source
git branch	git branch feature	List or create branches	-a, -d, -D
git checkout / switch	git switch -c feature	Change branches; switch is the modern branch-focused command	-b / -c
git merge	git merge feature	Combine another branch into the current one	--no-ff, --abort
git tag	git tag v1.0	Give a commit a version/name label	-a, -d, -l
git clone	git clone	Copy an existing remote repository locally	--depth, -b

3 • GIT'S THREE-TREE ARCHITECTURE

WORKING TREE	**INDEX / STAGING AREA**	**REPOSITORY / HEAD**
The files you are actively editing.	Changes selected to go into the next commit.	Saved commits that make up Git history.
git diff shows unstaged changes.	git diff --cached	git log

FLOW: Edit files → git add → changes enter the Index → git commit → snapshot enters the Repository. **Easy memory:** Working Tree = *what I have* • Index = *what I plan to save* • Repository = *what I saved*.

Quick checks: git status = what changed? • git diff = what changed but is not staged? • git diff --cached = what is staged? • git log --oneline = what commits exist?

References: Git documentation (git-scm.com/docs), Git User Manual / Reference, and Ubuntu command-line documentation. AI assistance was used in researching, organizing, wording, and designing this cheat sheet. For built-in help: git help <command> or man <command>.